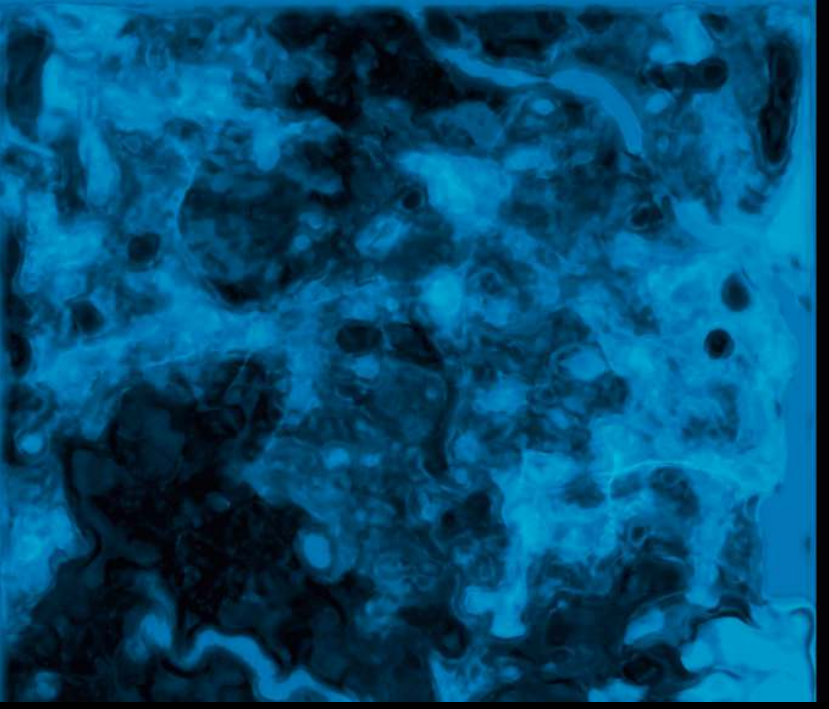
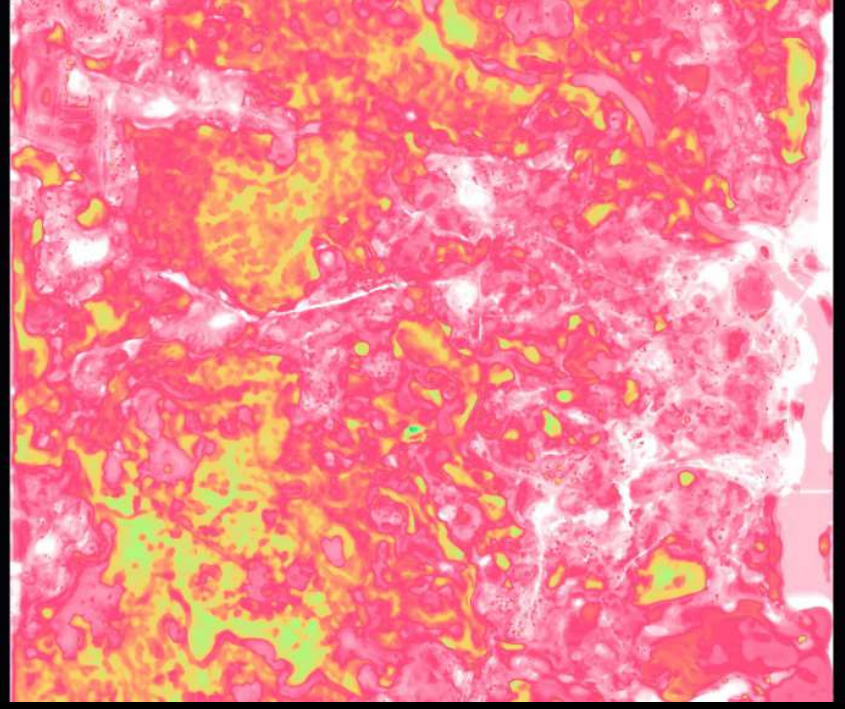
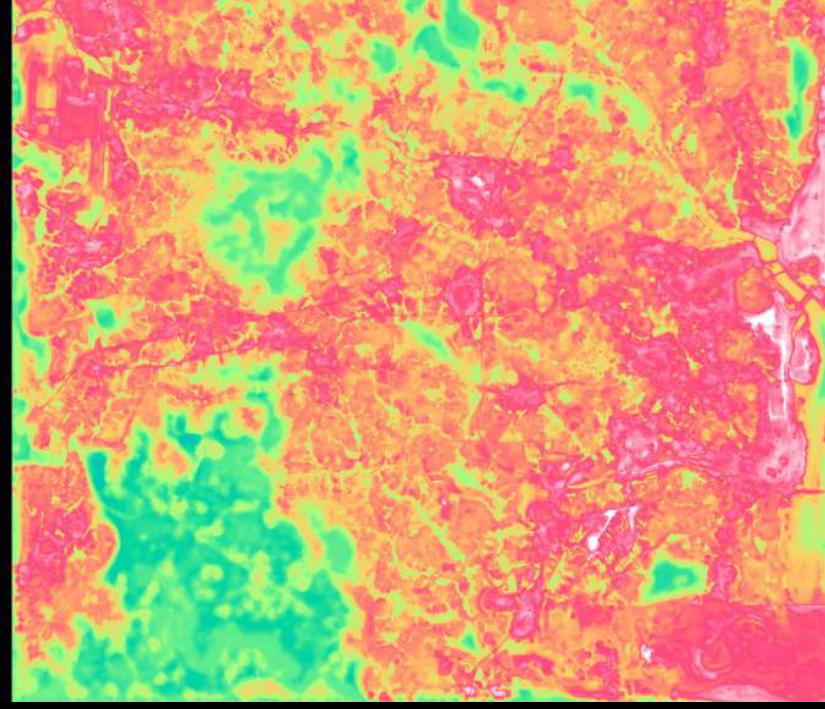




Modeling urban climates
Challenges and opportunities with AI

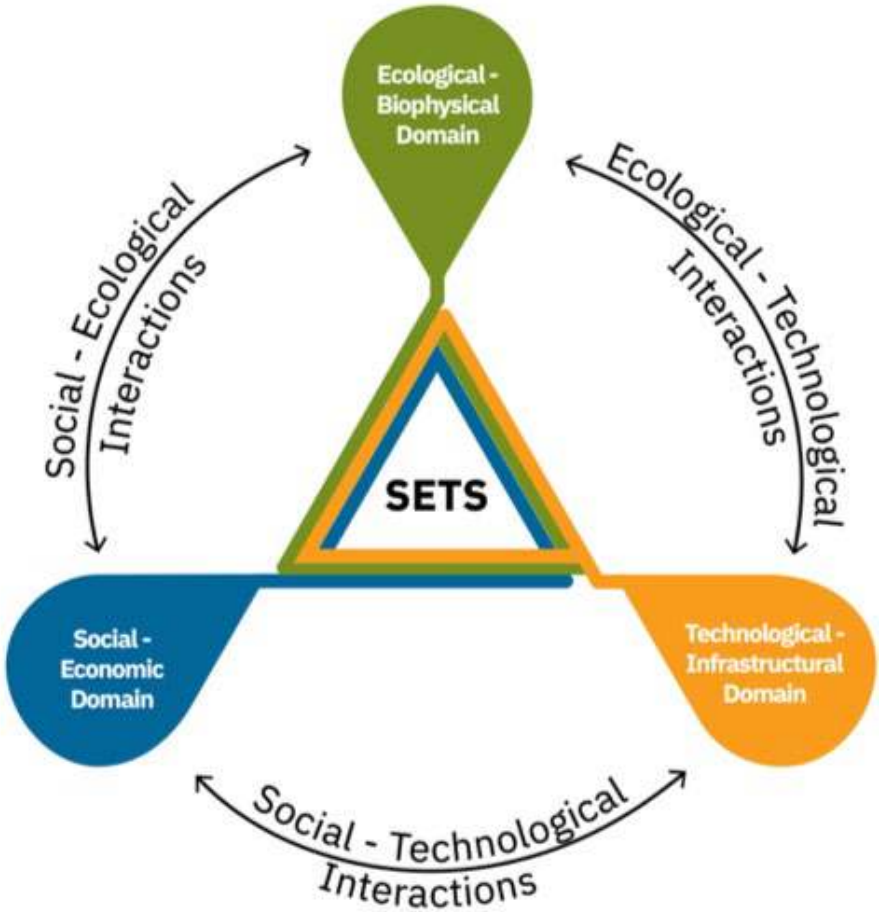
Luis E. Ortiz (with the help of many many others)
Assistant Professor of Climate Applications
(Acting) Virginia State Climatologist
Dept. Atmospheric, Oceanic, and Earth Sciences
George Mason University
Nov 5 2026

ClimateIQ

College of Science
VIRGINIA CLIMATE CENTER
George Mason University

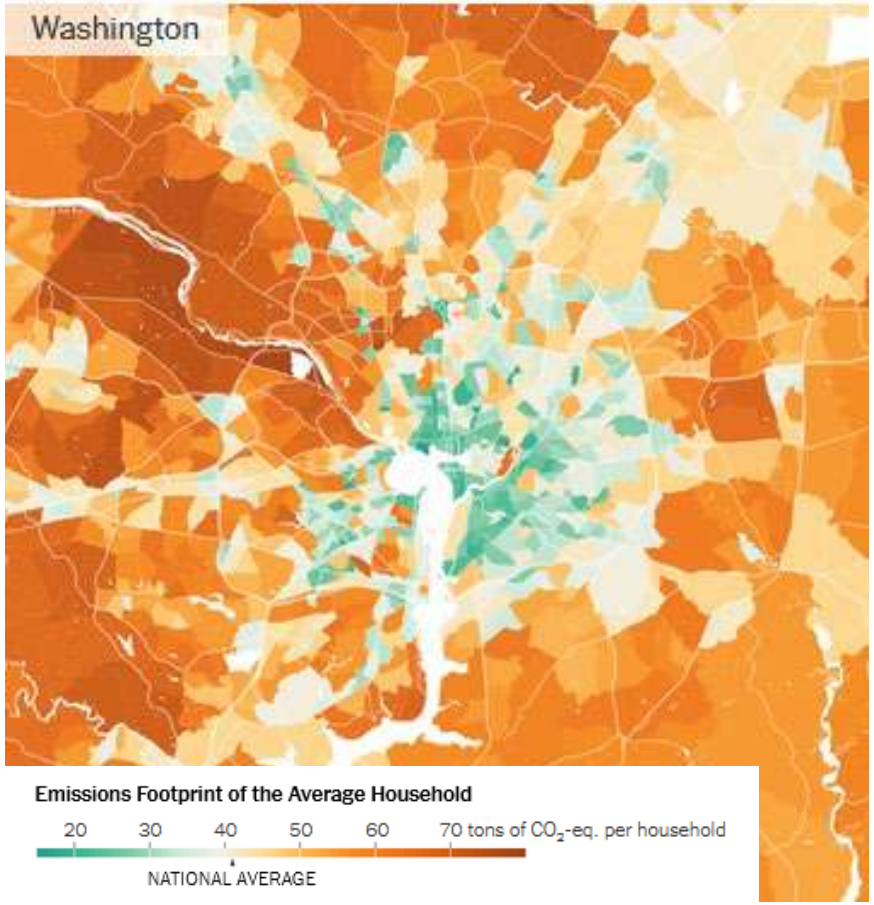
Why do we care about urban climates?

Cities are complex



Cities concentrate people, infrastructure, and

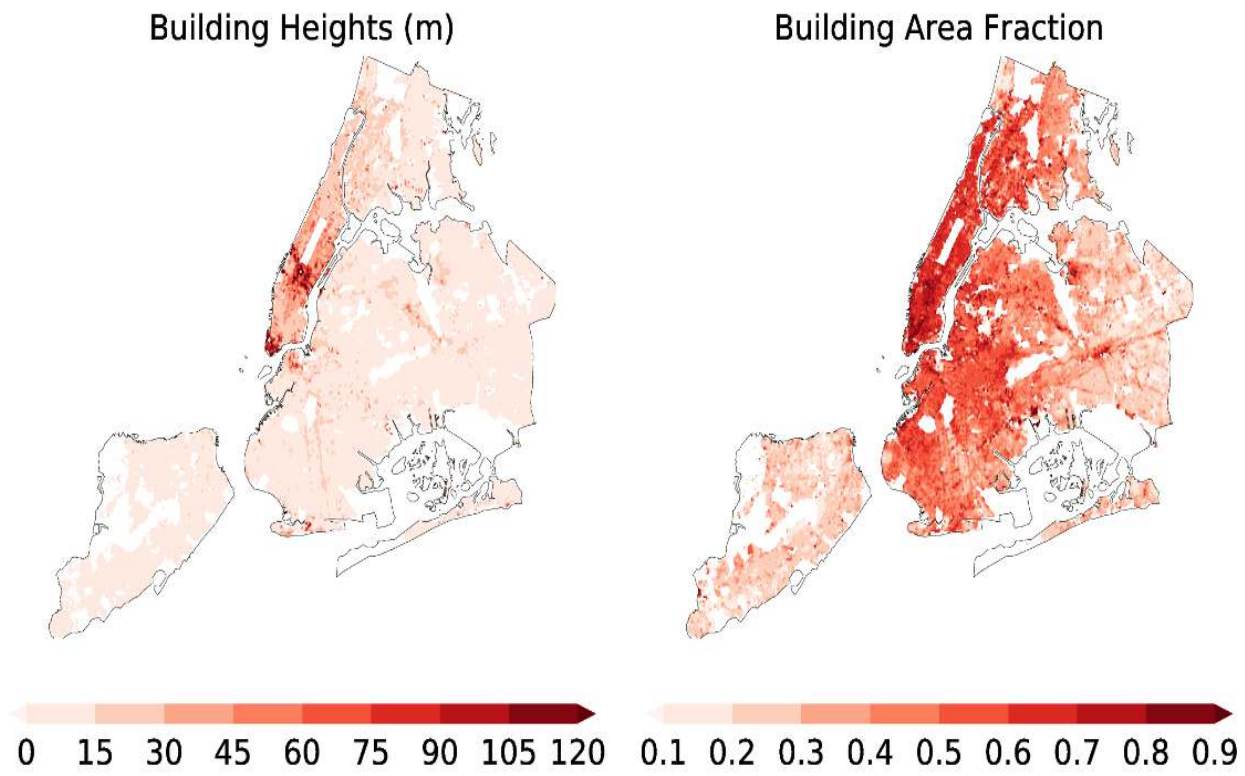
- Cities house over half of the global population (projected to reach nearly 70% by 2050) and generate over 80% of global GDP
- Urban areas consume roughly 75% to 80% of global energy and account for 67% to 72% of global greenhouse gas emissions*
- Opportunities for impactful climate action, trialing and scaling solutions, and more "bang for your buck"



Source: New York Times: **The Climate Impact of Your Neighborhood, Mapped,** Nadja Popovich, Mira Rojanasakul, and Brad Plumer

Cities affect local climates

- Modification of natural land cover
  - Built surface materials have VERY different thermal and aerodynamic properties to natural surfaces
  - Complex geometries
  - Impervious surfaces



“Thus, under the varying circumstances of different Sites, different Instruments, and different Positions of the latter, we find London always warmer than the country, the average excess of its temperature being 1.579° [deg F]”

- Luke Howard, The Climate of London (1833)

Cities modify the energy balance at the land surface

Net radiative flux
Incoming/outgoing solar

Sensible heat
Convection, advection

Ground conduction

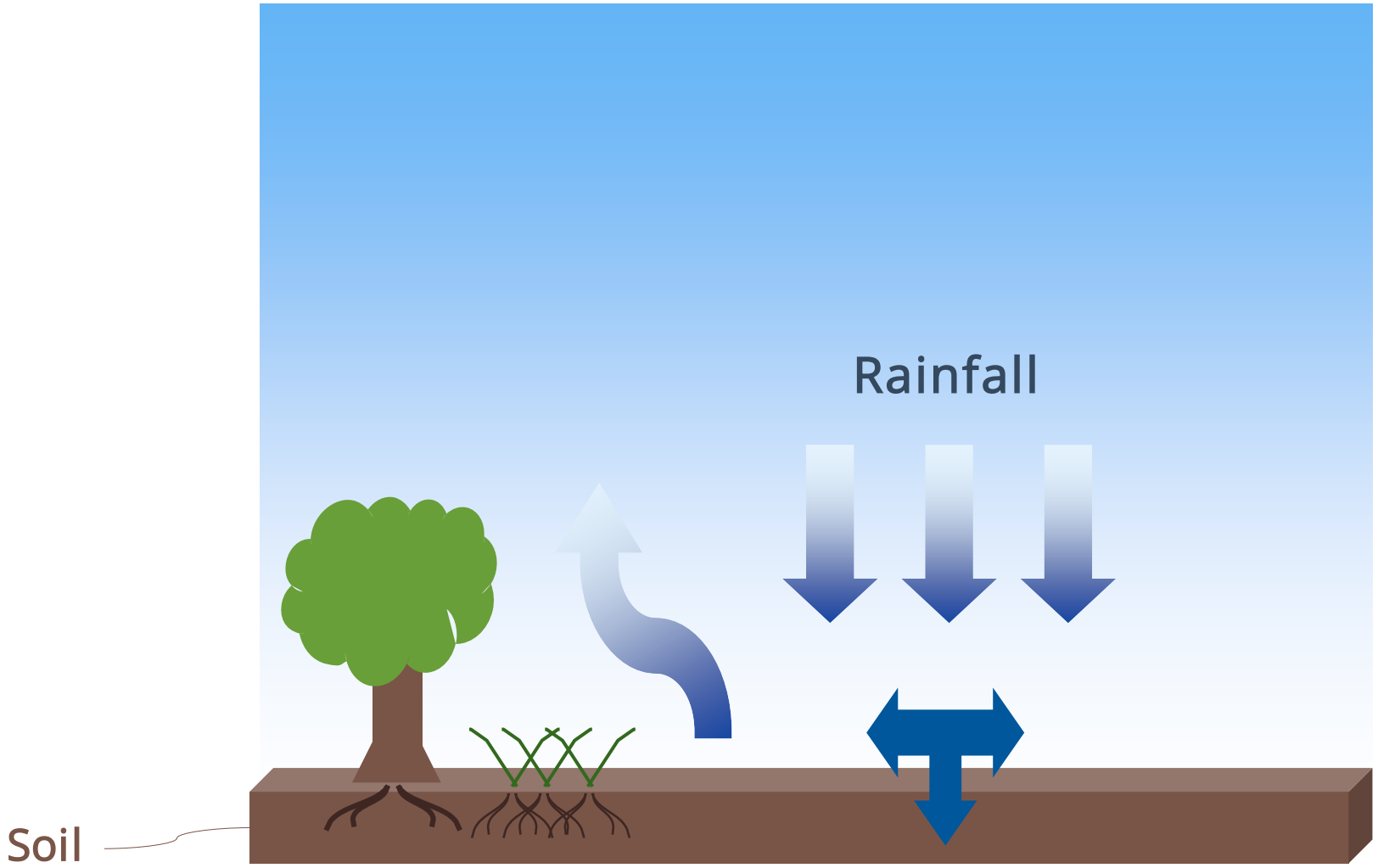
$$F_{net} + F_{storage} = F_{Hs} + F_{Es} + F_{Gs} + F_{anthropogenic}$$

Heat storage
Building materials
Increased volume

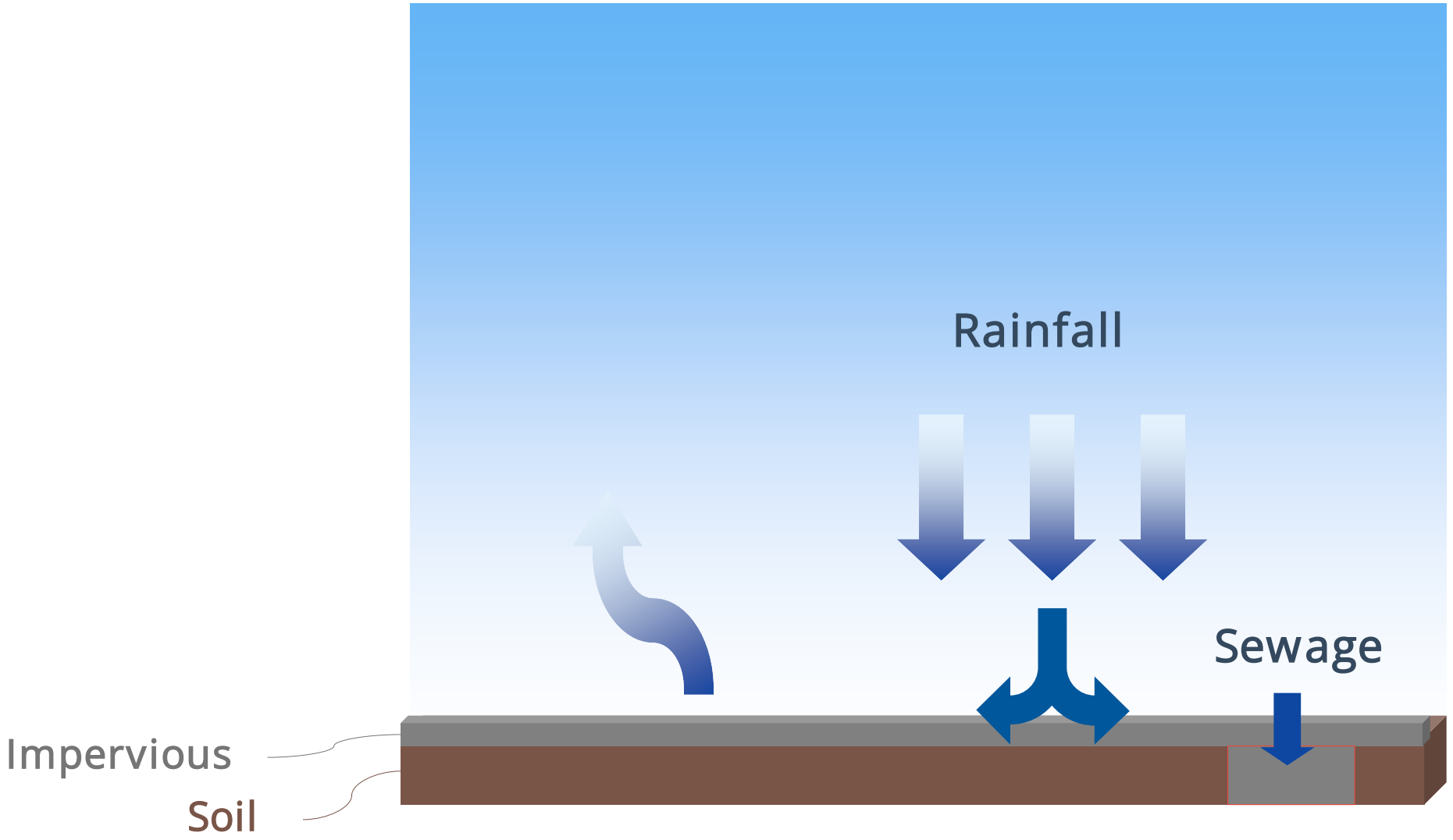
Latent heat
Evaporation, vegetation

Anthropogenic heat
Air conditioning
Traffic
Other sources

Water in non-impervious surfaces

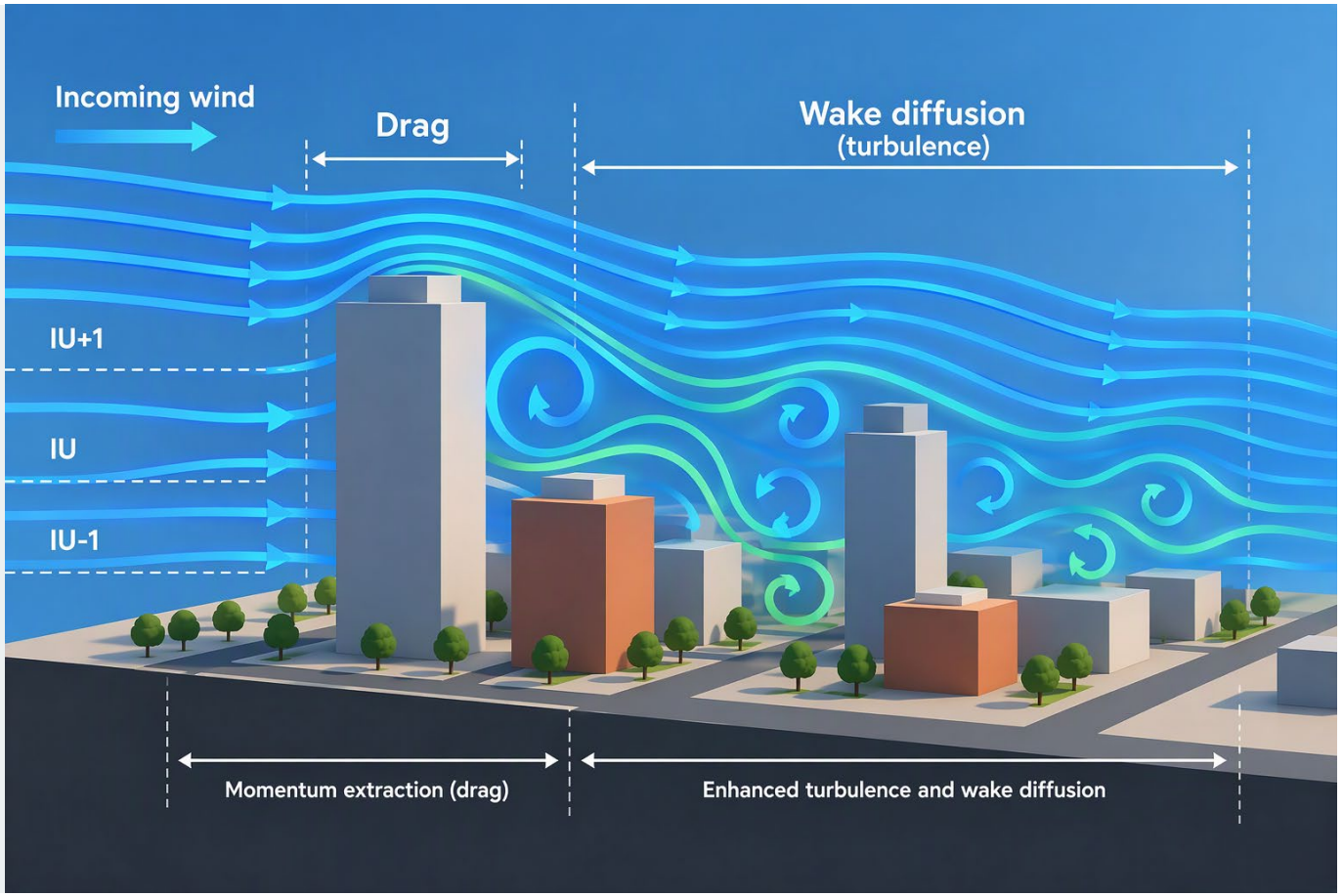


Water over impervious surfaces



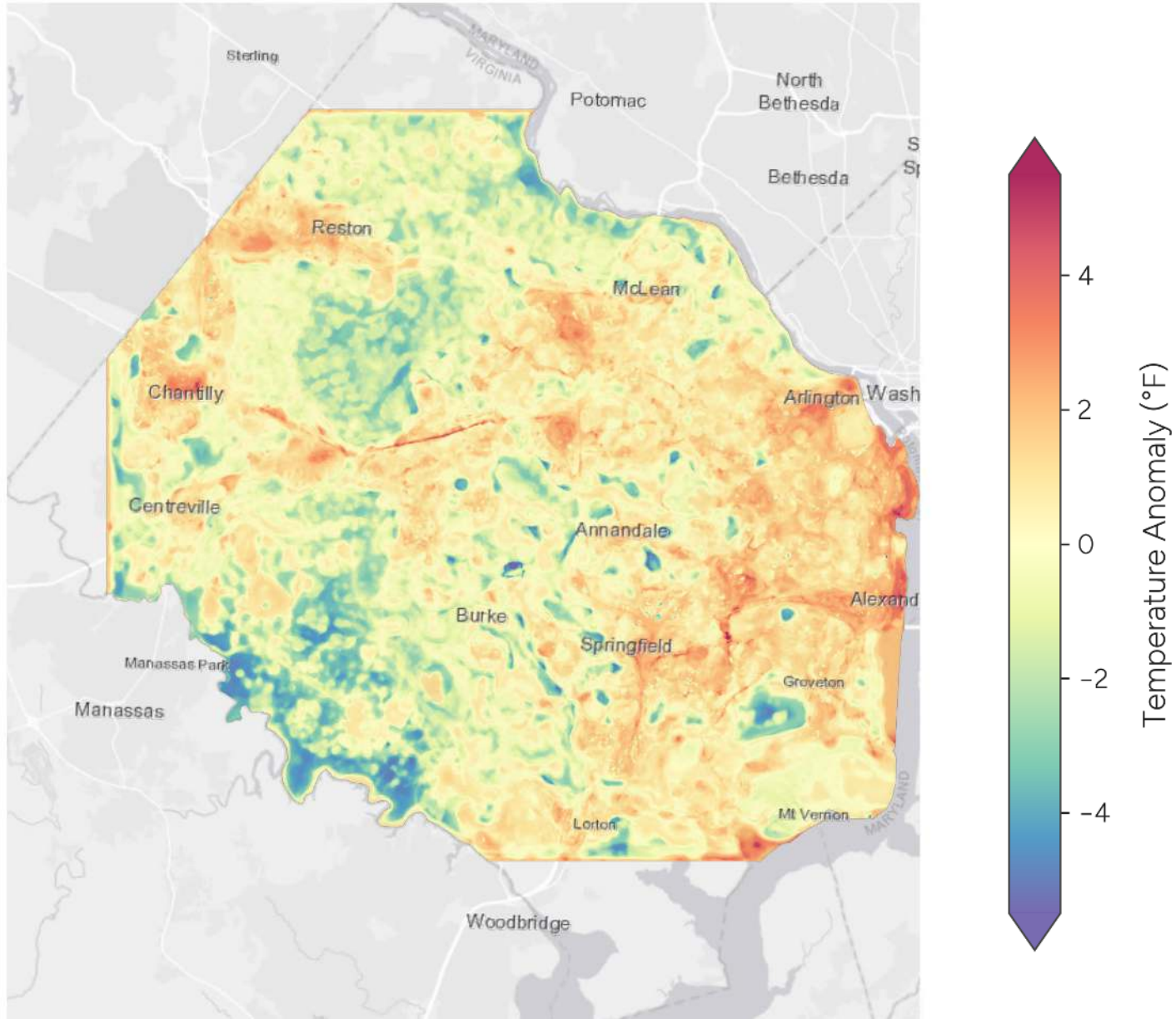
Cities and winds

- Drag: Buildings act as obstacles to wind flow, disrupting the smooth movement of air. This disruption creates areas of high pressure on the windward side (where the wind hits the building) and low pressure on the leeward side (the side sheltered from the wind), generating a force that opposes the movement of the air
- Turbulence: Buildings, with their various shapes and sizes, act as significant generators of turbulence. As wind encounters a building, it is forced to flow around, over, and sometimes through the structure. This disruption of the smooth airflow leads to the formation of turbulent eddies and wakes.



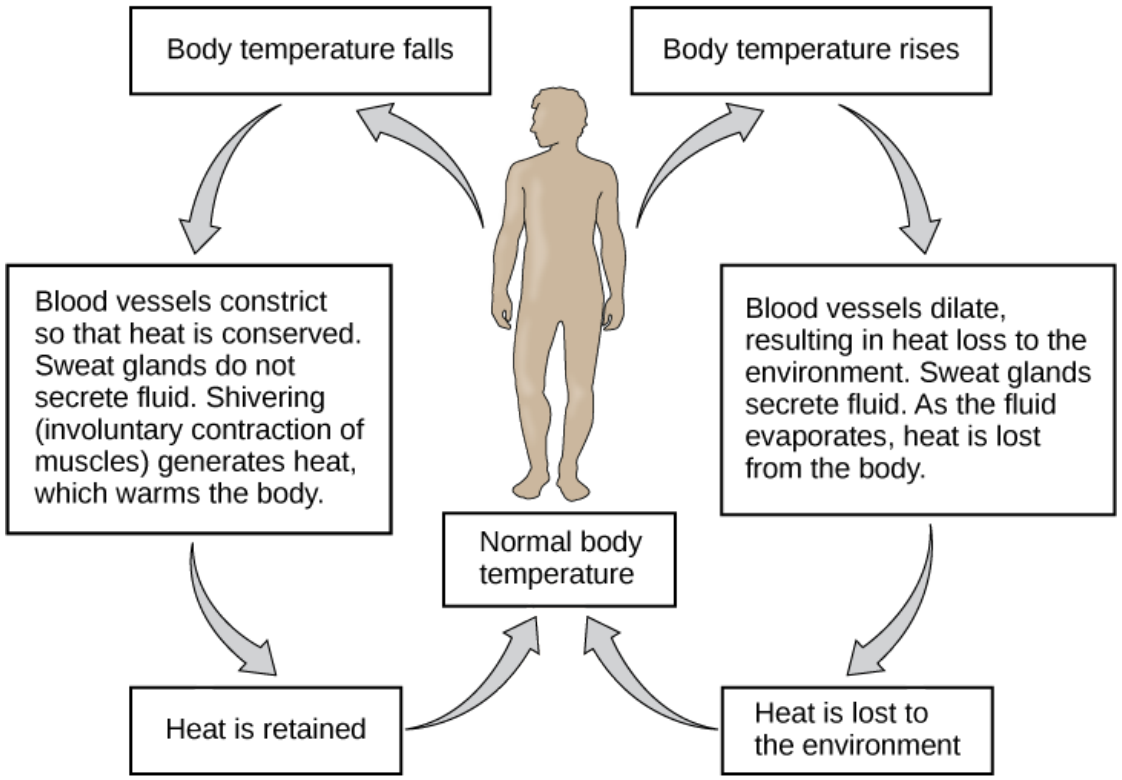
Afternoon (AF) — Spatial Temperature Anomaly

Deviation from Spatial Mean (95.3 °F) • Range: -12.7 °F to +7.6 °F



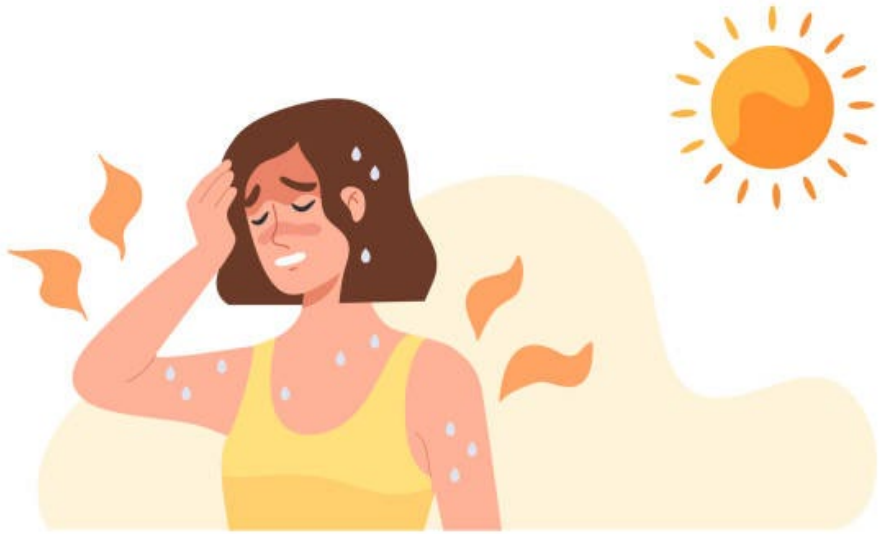
Human thermoregulation

- The human body can only operate within a narrow range of internal temperature. The process by which the body maintains this body temperature is called *thermoregulation.*
- When exposed to higher temperatures, the body initiates *cooling mechanisms:*
  - Transporting blood from internal organs to muscles and skin:
  - Activation of sweat glands:
  - Rapid respiration



Heat Strain

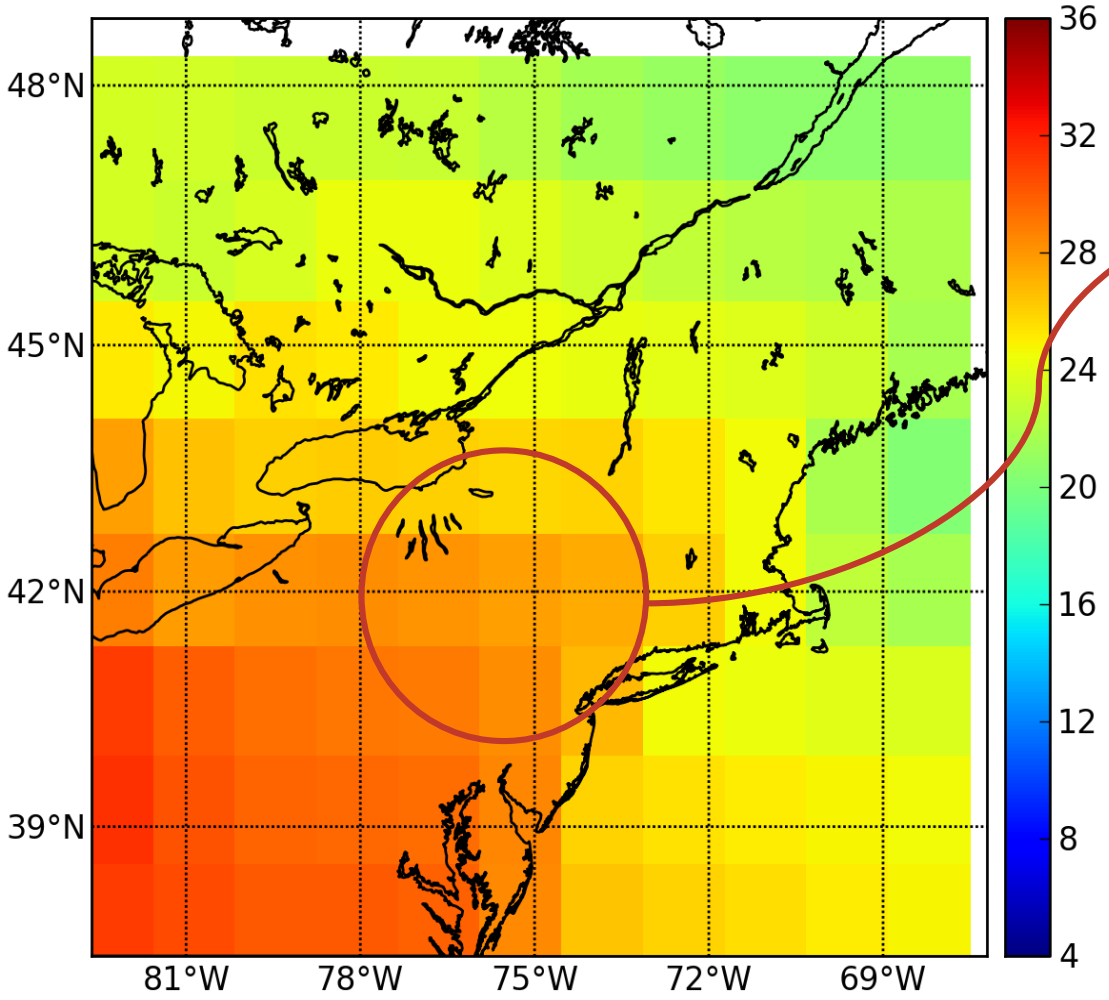
- Heat Strain: physiological and psychological responses of the body to heat stress, which include measurable changes in various bodily functions:
  - including heart rate,
  - core body temperature,
  - skin temperature, and
  - sweating rate
- During extreme heat and high humidity, the body's ability to cool itself is significantly impeded. Extreme heat reduces the dissipation of heat from the body's surface, while high humidity slows down the evaporation of sweat, the body's primary cooling mechanism



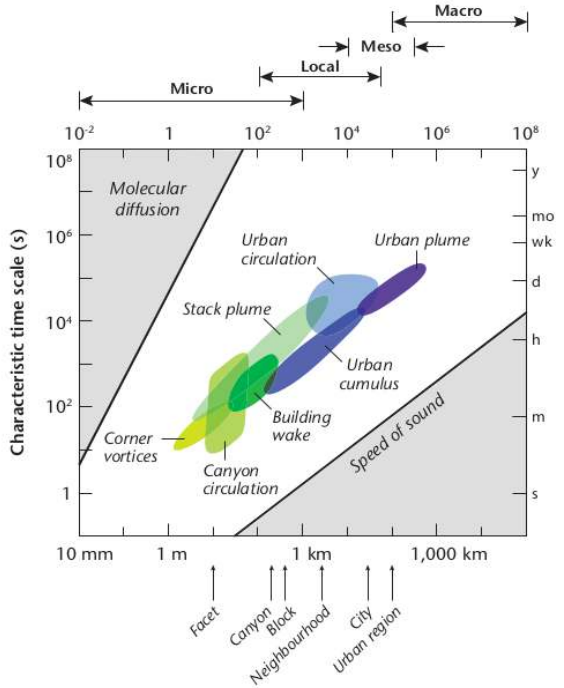
Challenges in urban climate

Traditional weather and climate prediction

Climate models and the scale problem



Climate models provide information at scales of ~25-100 km*



Urbanized NWP

- Developed by NOAA and consortium of academic and federal groups
  - Solves fully compressible, Euler non-hydrostatic governing equations
    - Modified 3rd order Runge-Kutta time integration (RK3- scheme, Wicker and Skamarock 2002)
- Urban Parameterization
  - Urban-specific processes, radiative transfer, land surface (evaporation from soils, plants, etc), moisture transport (both phase change and via convection/advection)
  - Building Effect Parameterization (Martilli et al 2002) + Building Energy Model (Salamanca et al (2010)

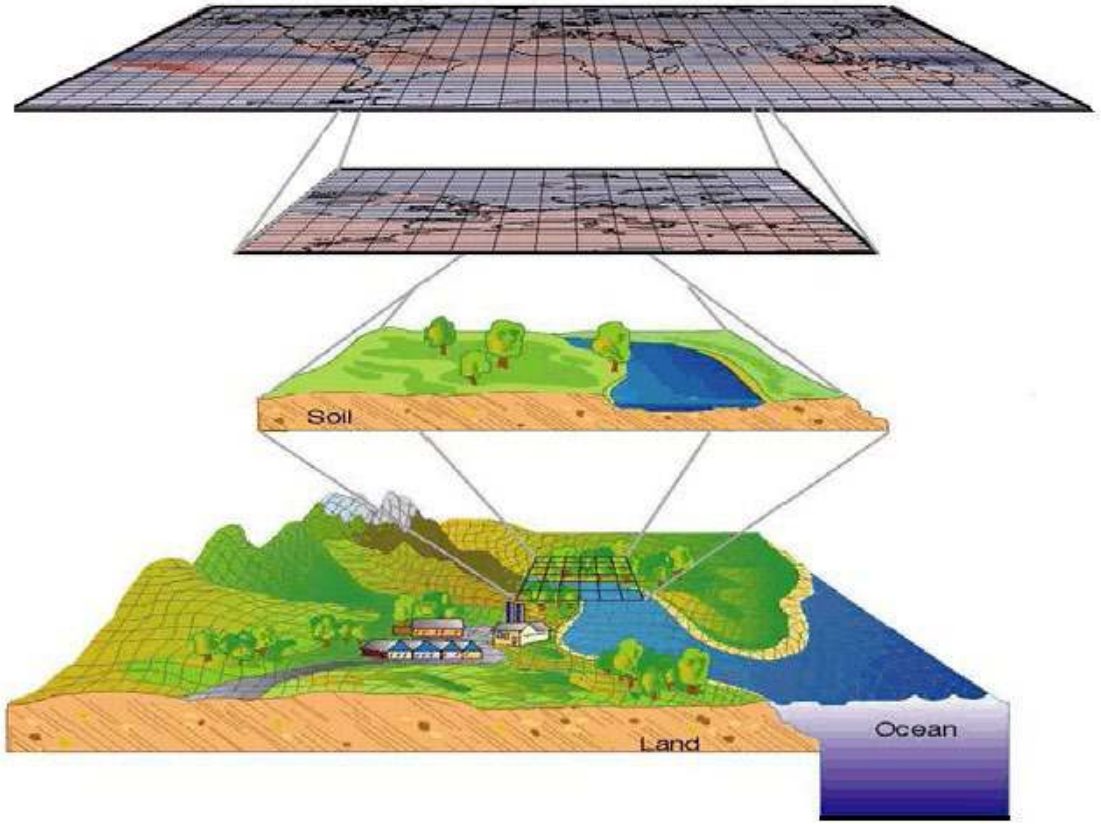
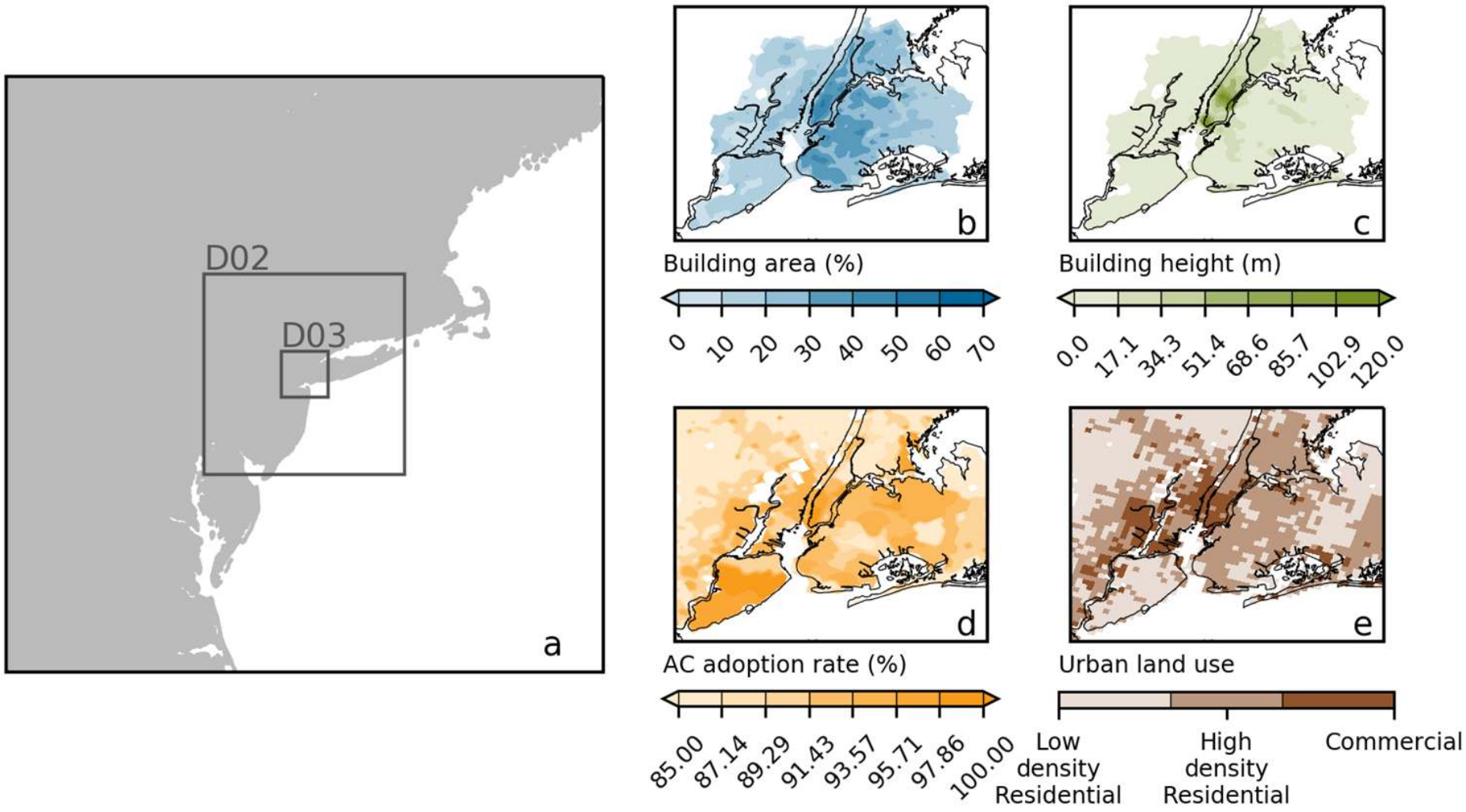
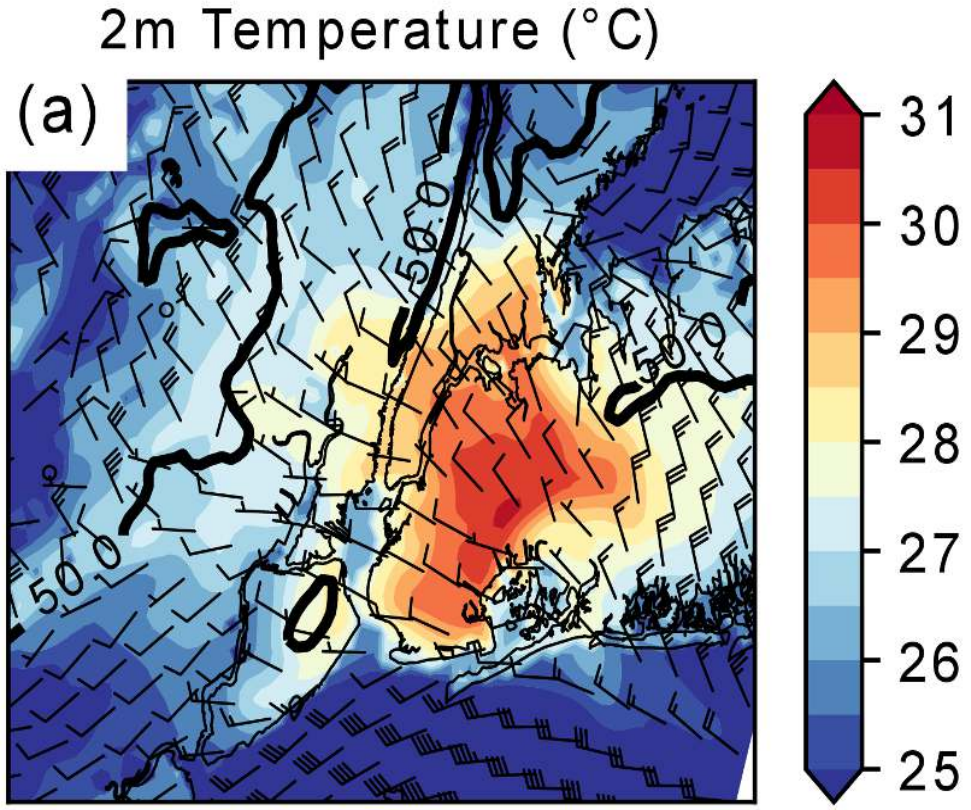


Image Reproduced from Khan and Pilz (2018) under a Creative Commons License. Khan and Pilz (2018)

Urban parameters



Dynamical downscaling



Using coarse global model data to drive finer scale regional models

NYC is easy: data is available from multiple sources, and we have a lot of previous studies to compare against.

L. Ortiz, J. E. González, R. Horton, W. Lin, W. Wu, P. Ramamurthy, M. Arend, and R. Bornstein, High resolution projections of extreme heat in New York City. *Int. J. Clim.*

NCAR FastEddy: Downtown Dallas

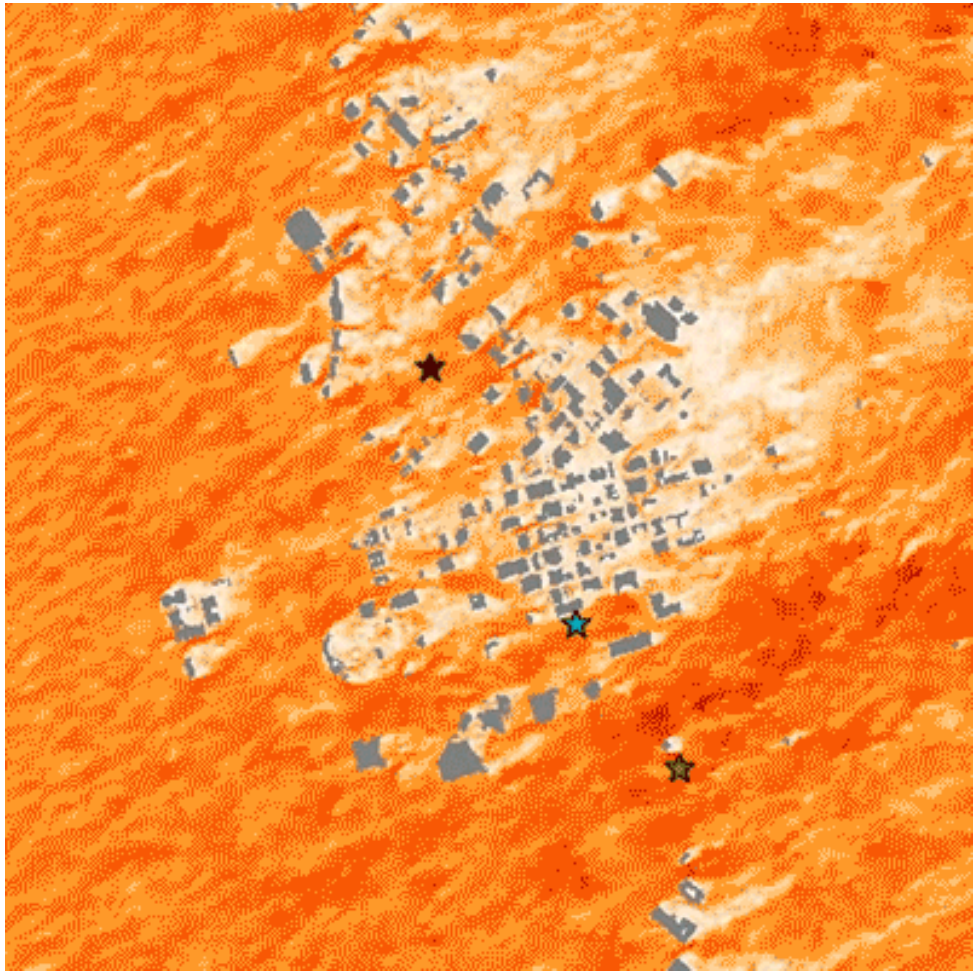
NCAR FastEddy model

Buildings explicit in model

5 m resolution

Runs on GPUs

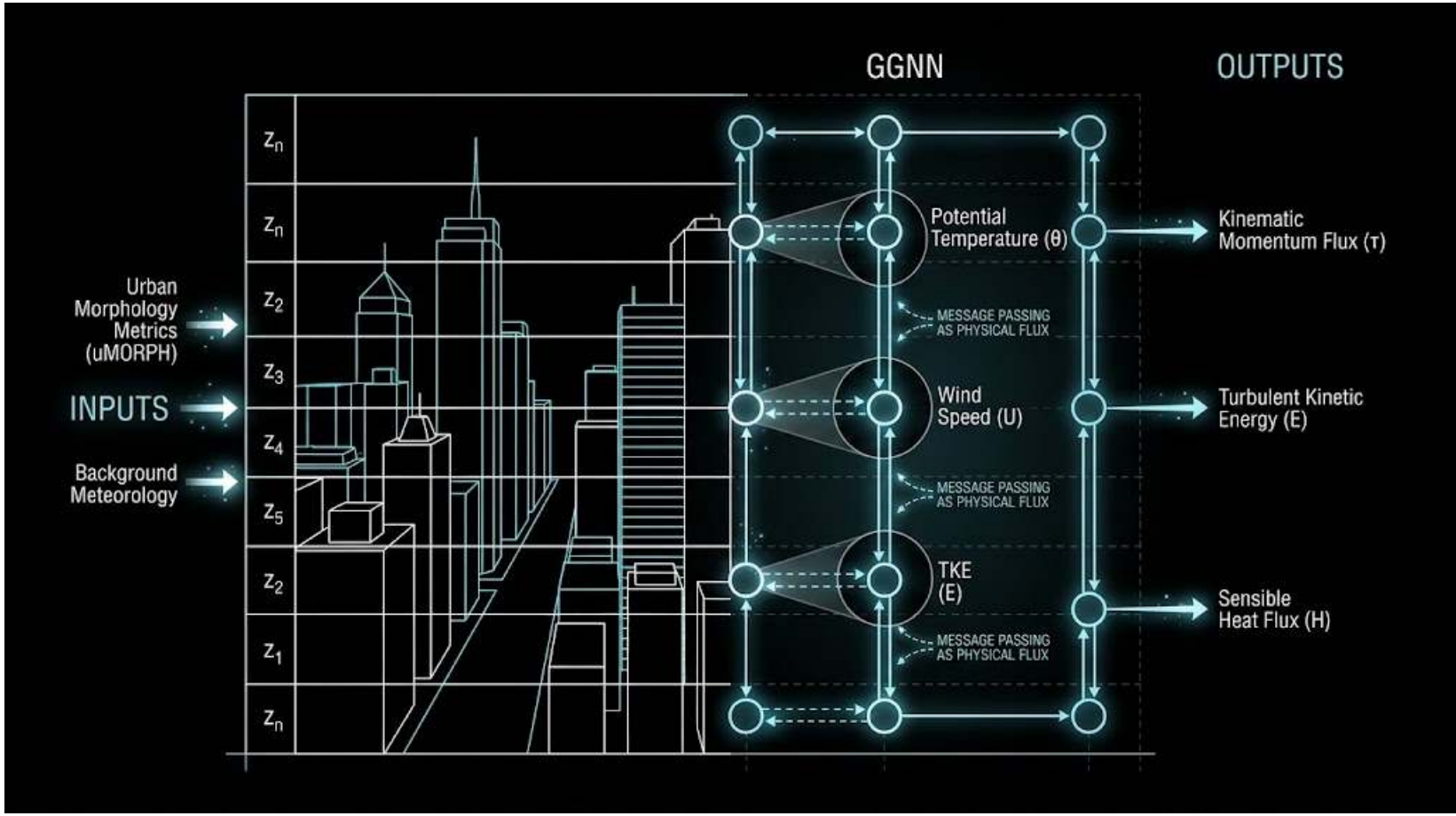
Total simulated time: 45min-3hours (up top 8 hour simulation cost)



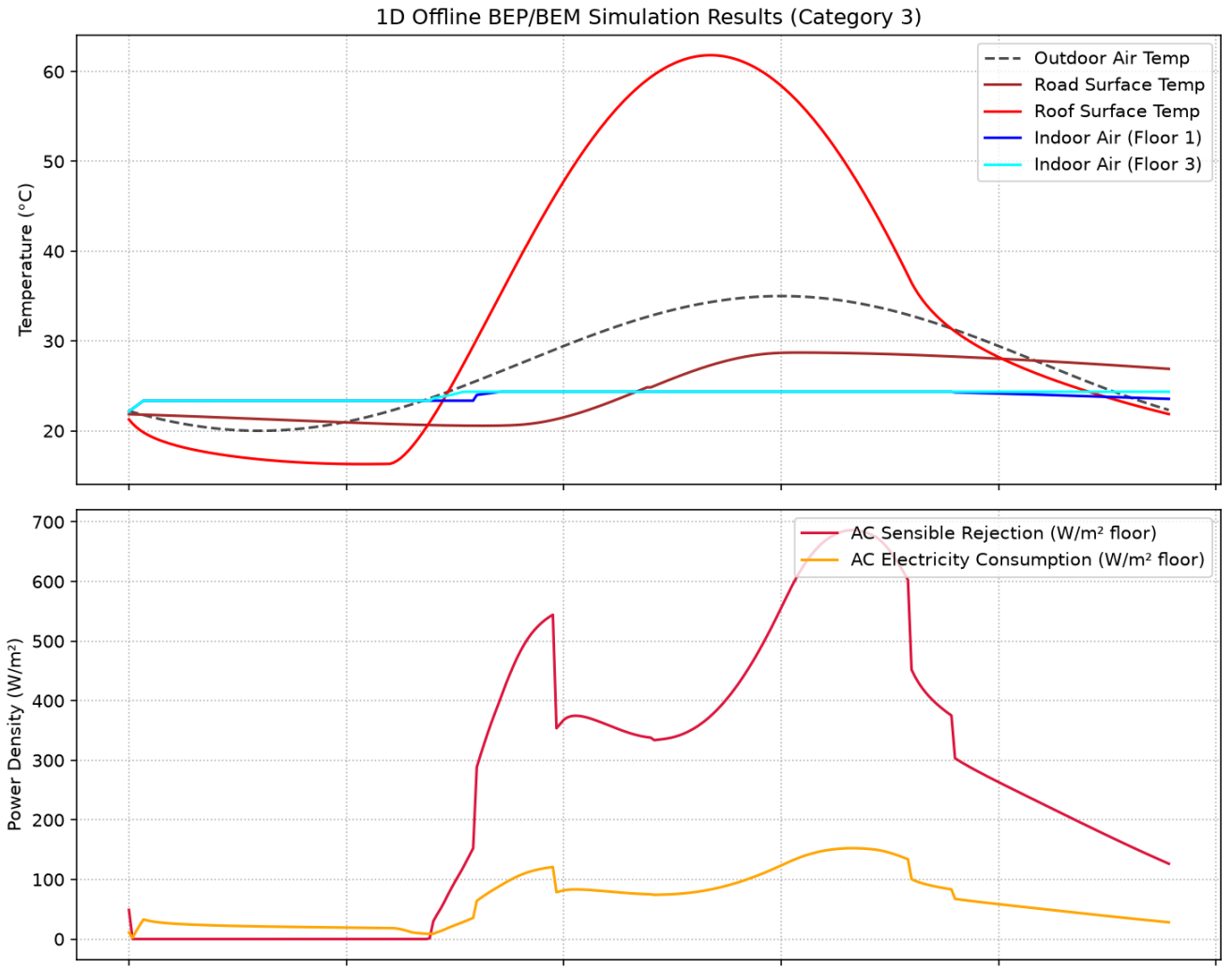
Source: https://ral.ucar.edu/fasteddy
Recently open sourced!

Opportunities in AI

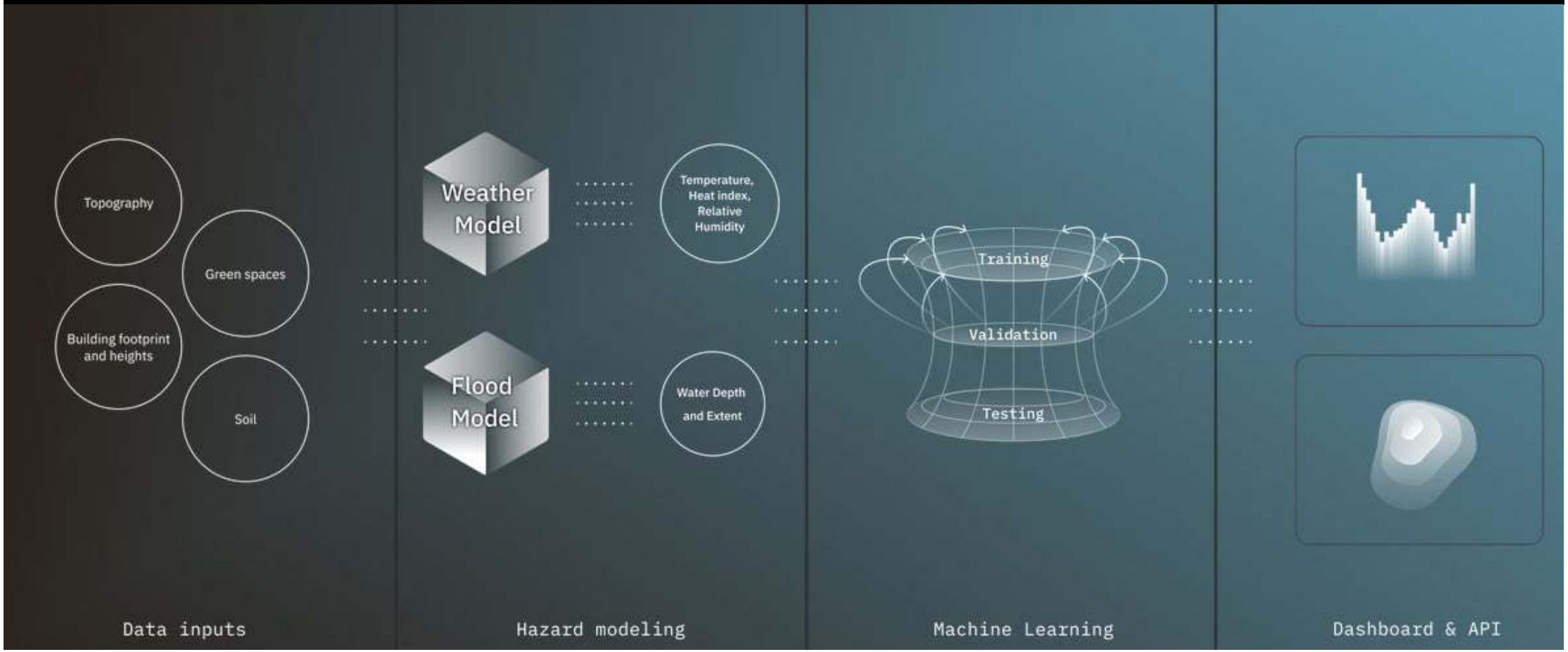
- Computational costs are *the* limiting factor in urban climate modeling.
- Advances in AI for weather and climate hard to adopt: they rely on large-scale, but coarse data for training (~30km).
- Cyborg models: Substitute slow components of physics based model with AI-based emulators?
- Eat the cost and create a training library for Urban AI climate models?



Emulating offline urban physics

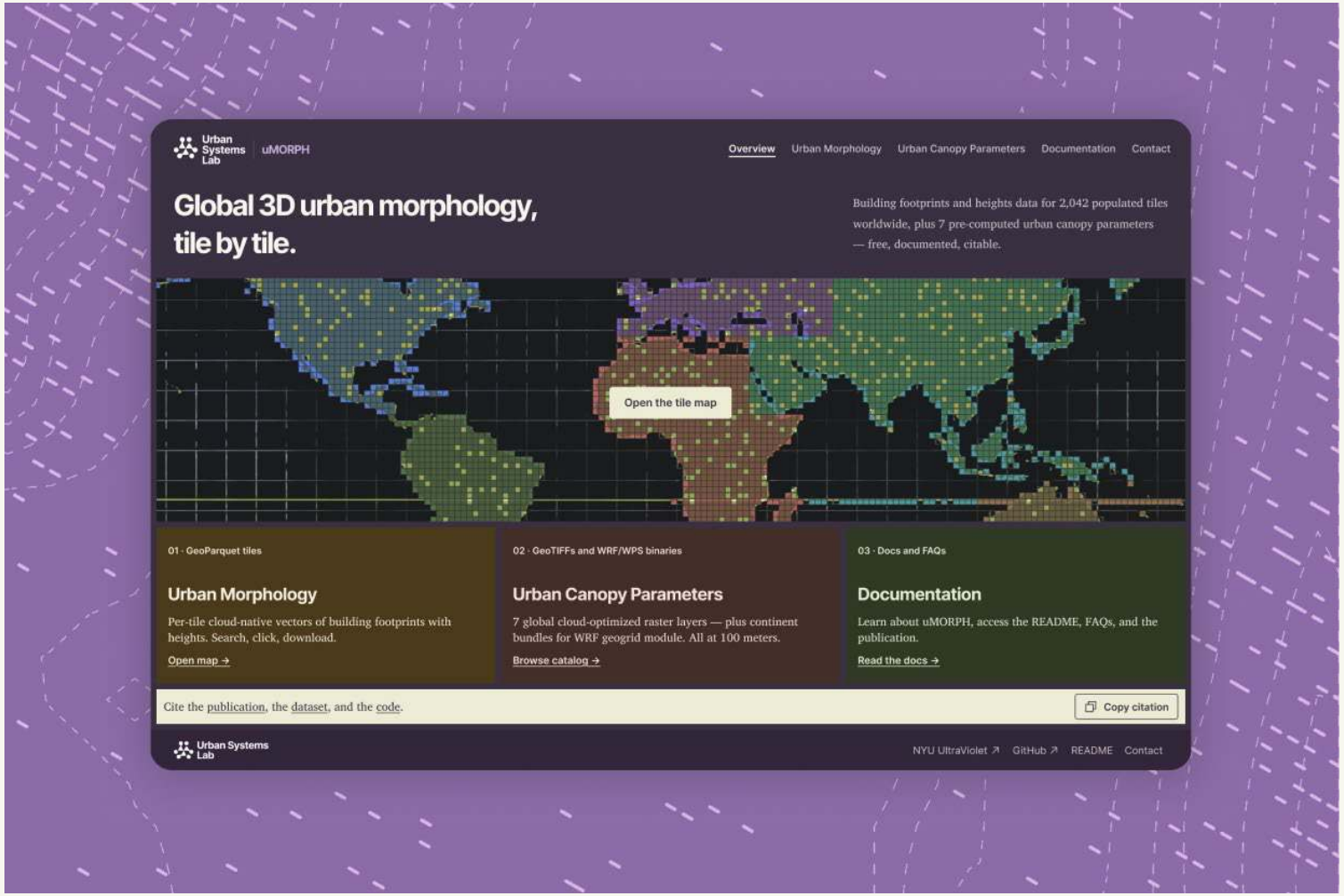


Eating the cost (pay once cry once?)



Urban Systems Lab

uMORPH



Urban Morphology - uMORPH

uMORPH is a open-access global dataset of Level-of-Detail-1 building footprints with heights and 100-meter Urban Canopy Parameters with coverage representing 7.84 billion people across all continents.

Designed for easy integration and faster rendering in workflows.

Supporting neighborhood-scale urban climate modeling, machine-learning applications, and data visualization.

Partners & Collaborators

GEORGE MASON UNIVERSITY

NYU | UltraViolet

TORCH

with support from Google.org



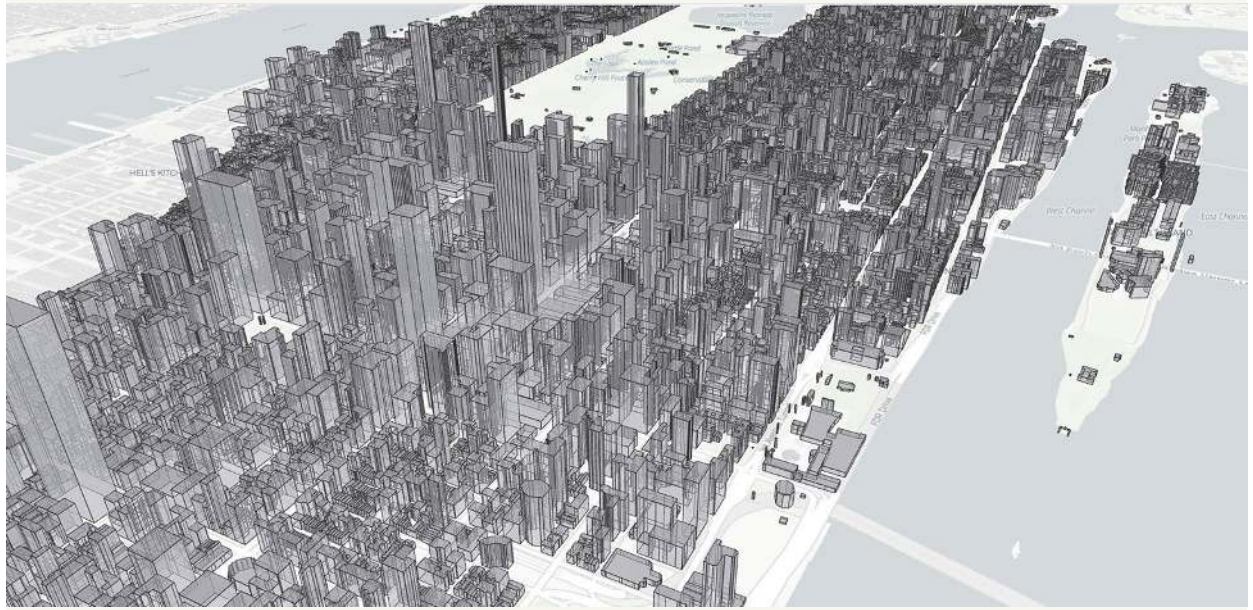
Urban Systems Lab

Footprints & heights: ~2.38 billion polygons

Urban Canopy Parameters: 7 (100-meters)

Cities: 98.4% with 50,000+ people

Population: 7.84 billion (94.5% global)



Urban Systems Lab | uMORPH

Overview | Urban Morphology | Urban Canopy Parameters | Documentation | Contact

Urban Canopy Parameters

7 urban canopy parameters (24 gridded layers in total) derived from LOD1 building footprints and heights. Distributed as 100-meter global Cloud-Optimized GeoTIFFs (download per layer) — and as 100-meter WRF/WPS geogrid binaries (download per continent).

Sizes are archived. Unarchived sized may be substantially larger.

View metadata on Globus →

#	Layers (cloud-optimized GeoTIFFs)	Unit	UCM Schemes	Size	
01	Mean Building Height	meter	SLUCM	4.6 GB	⬇
02	Area-Weighted Mean Building Height	meter	SLUCM, BEP, BEP + BEM	4.6 GB	⬇
03	Standard Deviation of Building Height	meter	SLUCM	4.5 GB	⬇
04	Plan Area Fraction	dimensionless [0, 1]	SLUCM, BEP, BEP + BEM	5.1 GB	⬇
05	Building Surface-to-Plan Area Ratio	dimensionless	SLUCM, BEP, BEP + BEM	5.1 GB	⬇
06	Frontal Area Index (4 layers; 0,45,90,135 degrees)	dimensionless	SLUCM	21 GB	⬇
07	Building Height Distribution (15 layers; 5 m bins)	dimensionless fraction [0, 1]	BEP, BEP + BEM	23 GB	⬇

WRF geogrid binaries

Africa 2.4GB ⬇	Europe 2.4GB ⬇	South America 1.2GB ⬇
Asia 4.8GB ⬇	North America 2.4GB ⬇	Oceania 2.4GB ⬇

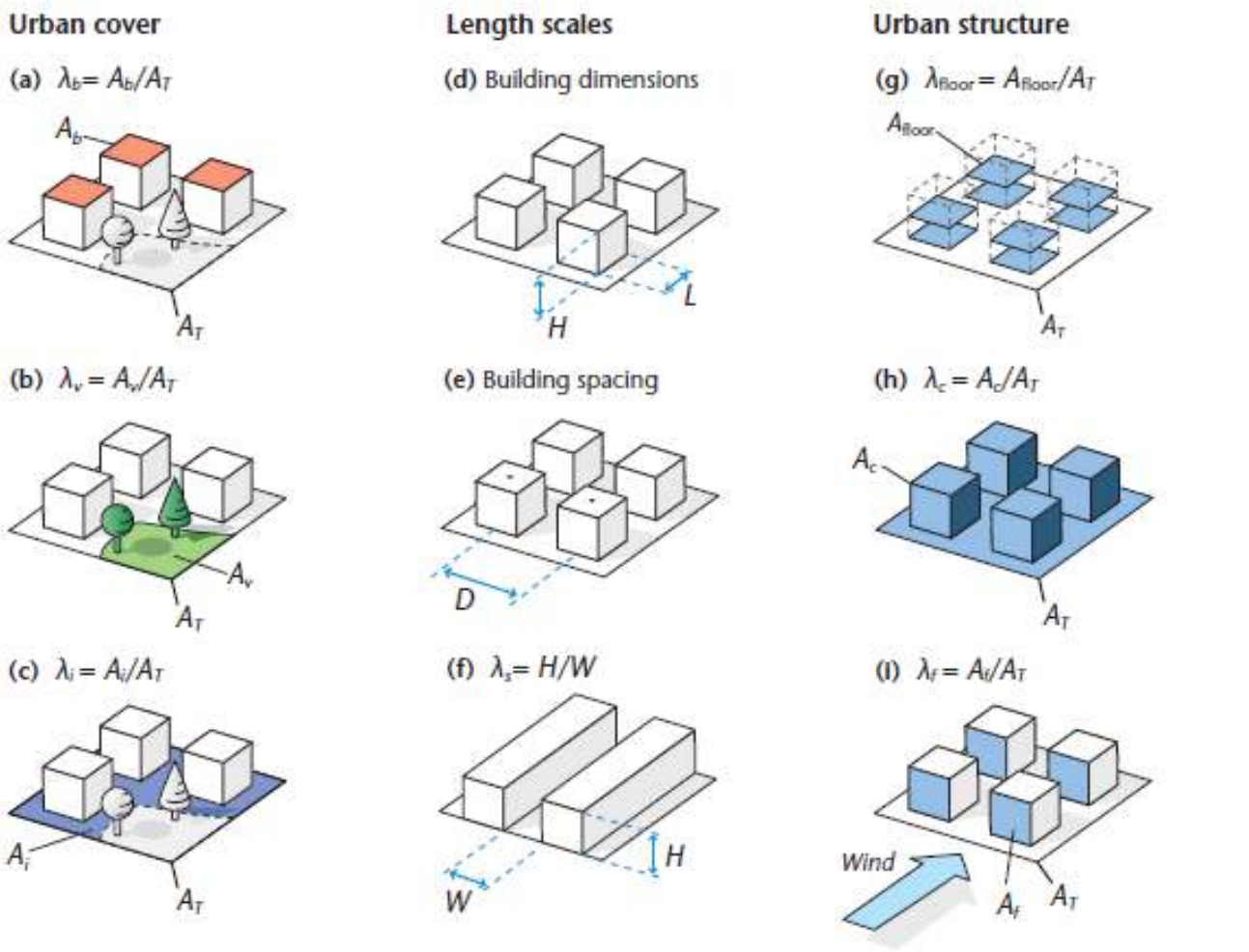
Urban Systems Lab

NYU UltraViolet ↗ | GitHub ↗ | README | Contact

What are urban canopy parameters (UCPs)?

UCPs help us translate descriptions of the urban environment **from geometric surfaces to a grid of values**

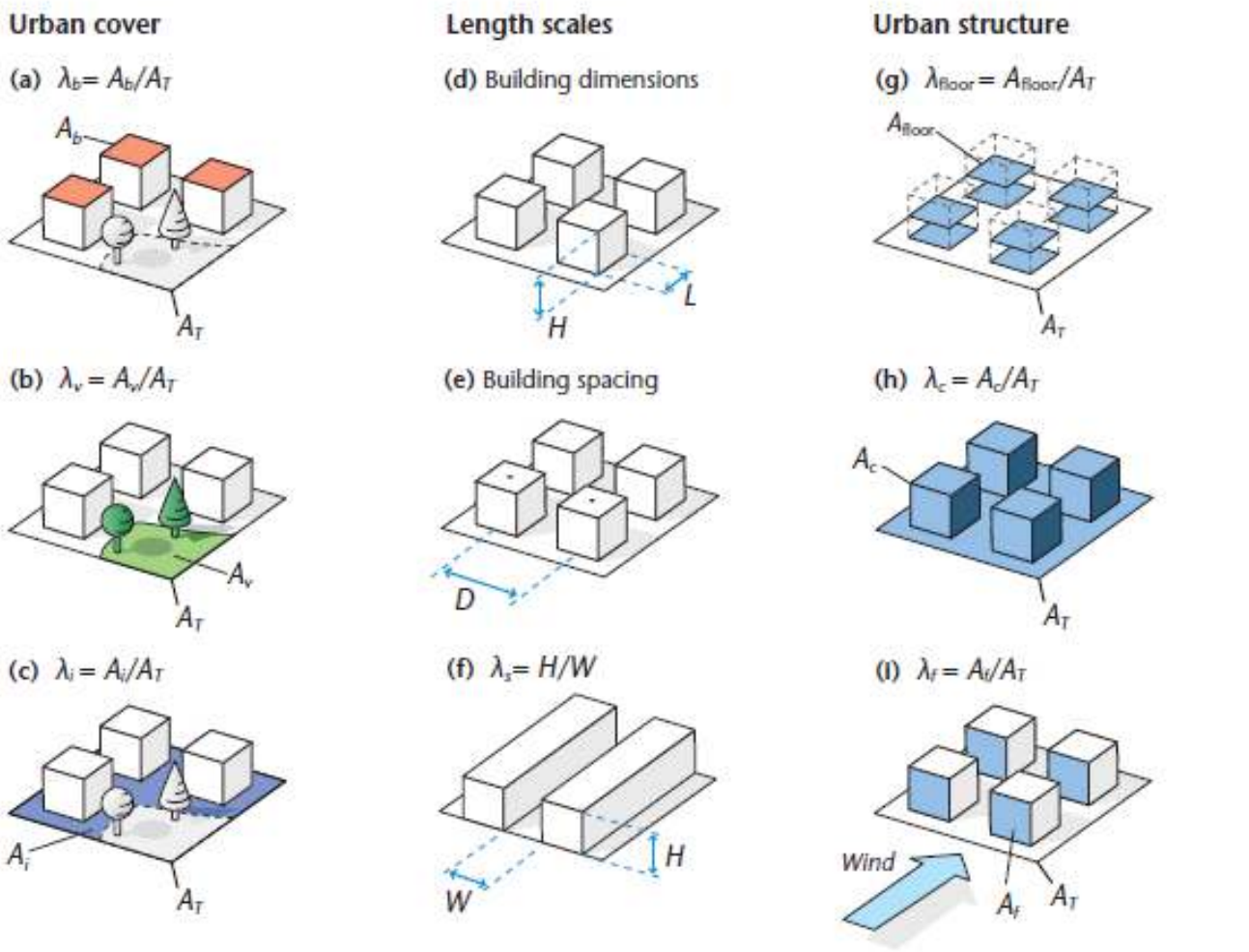
- Average building height
- Building height to street width ratio
- Building footprint to total area ratio



Why are urban canopy parameters (UCPs)?

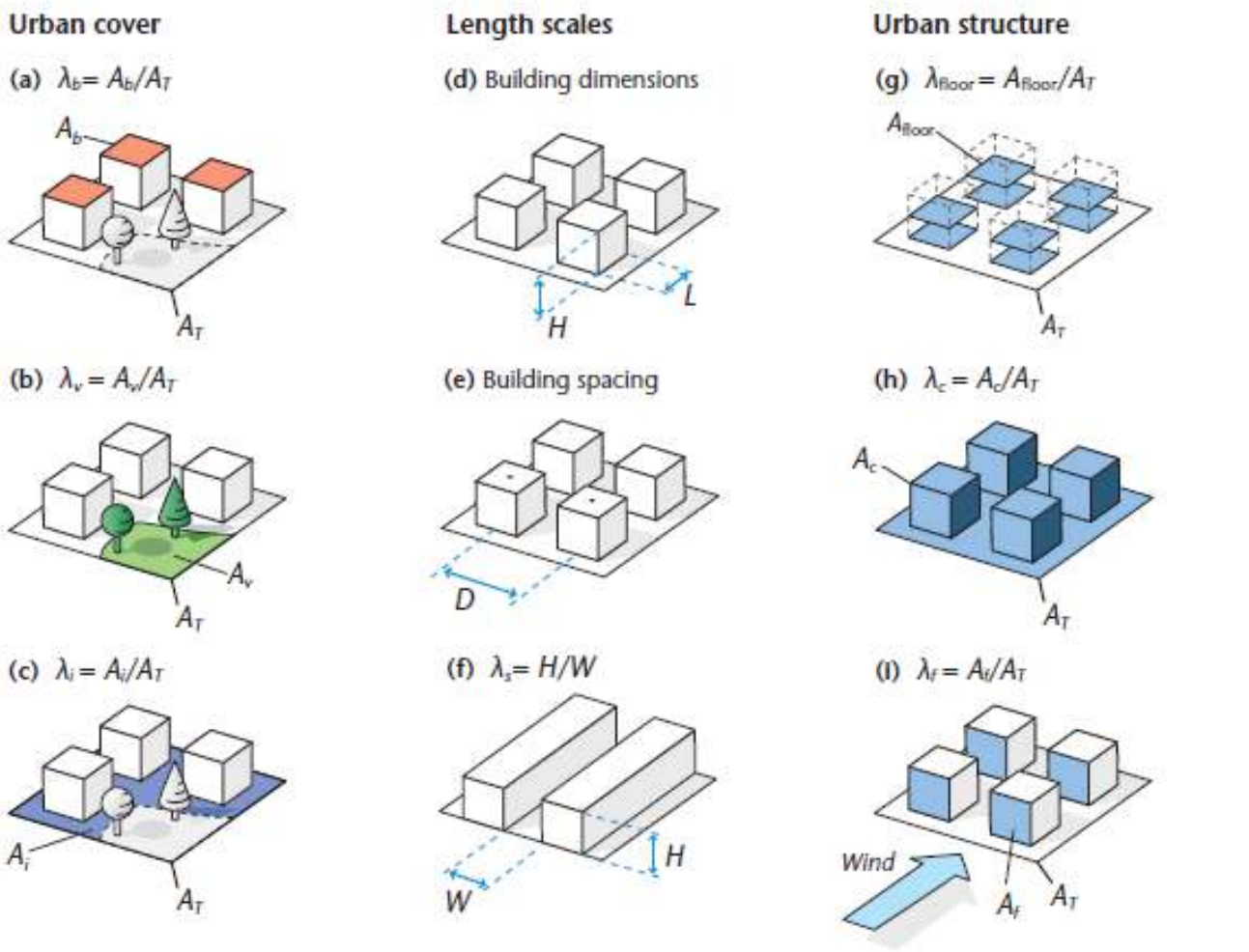
In numerical weather prediction, **buildings are smaller than typical grid sizes, but are important contributors to local weather:**

- Canopy urban heat islands
- Local transport processes
- Urban-induced thunderstorms

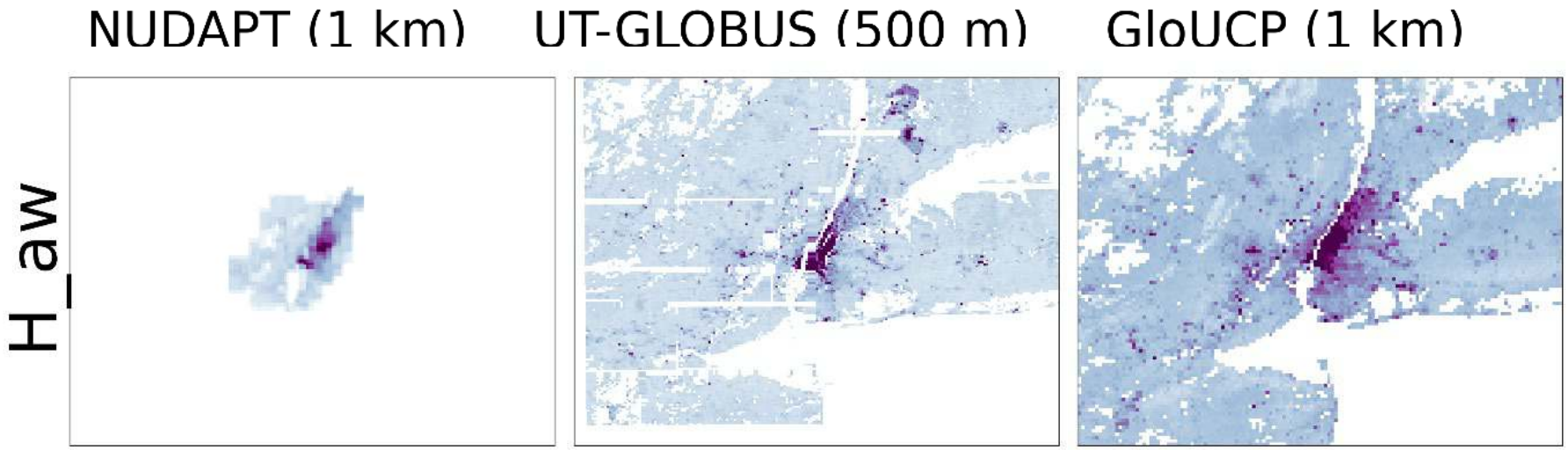


Why are urban canopy parameters (UCPs)?

To account for the physics of the urban environment**, we use UCPs to estimate the aggregate effect** of buildings and other urban cover on surface processes.

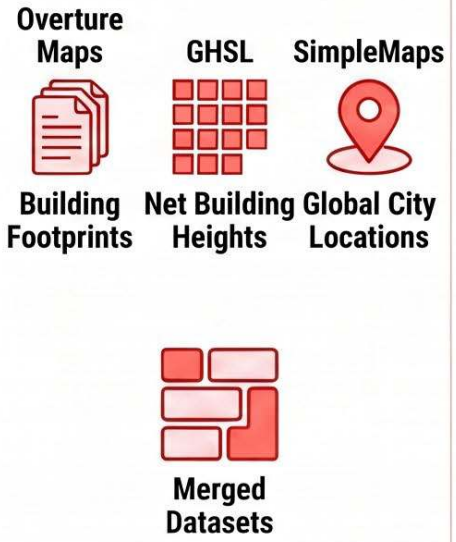


There are no truly global, current, unproblematic, or fine-scale UCPs available for public use



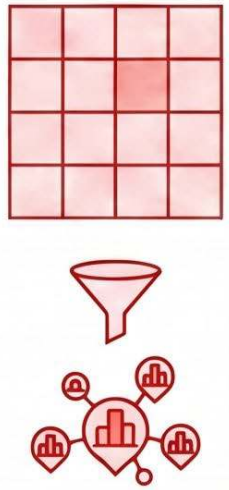
Introducing uMORPH

1. Data Integration



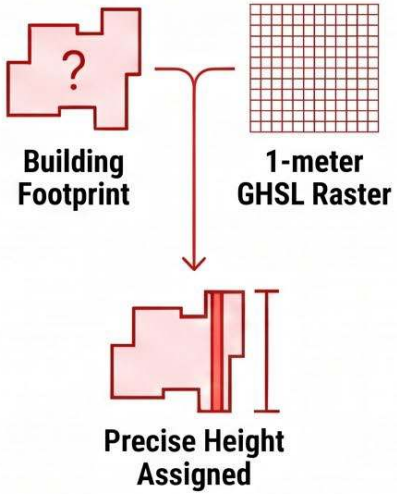
Workflow begins by merging three core open-source datasets for footprints, net heights, and global city locations.

2. Geographic Tiling & Filtering



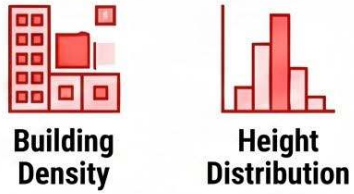
Earth divided into 2" × 2" tiles, filtering for over 11,000 urban centers with population > 50,000.

3. Height Infilling



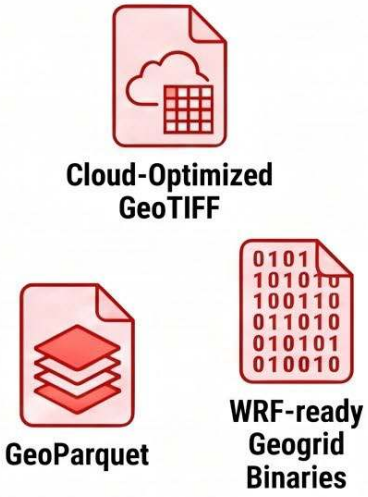
Footprints lacking height data intersect with a 1-meter resampled GHSL raster to assign height attributes.

4. Parameter Derivation



Computes 7 key morphological parameters (e.g., density, height distribution) across 24 unique data layers at 100-meter resolution.

5. Multi-Format Distribution



Final outputs converted into rasters, vectors, and binaries for immediate scientific use.

Global Population Coverage:
94.5%

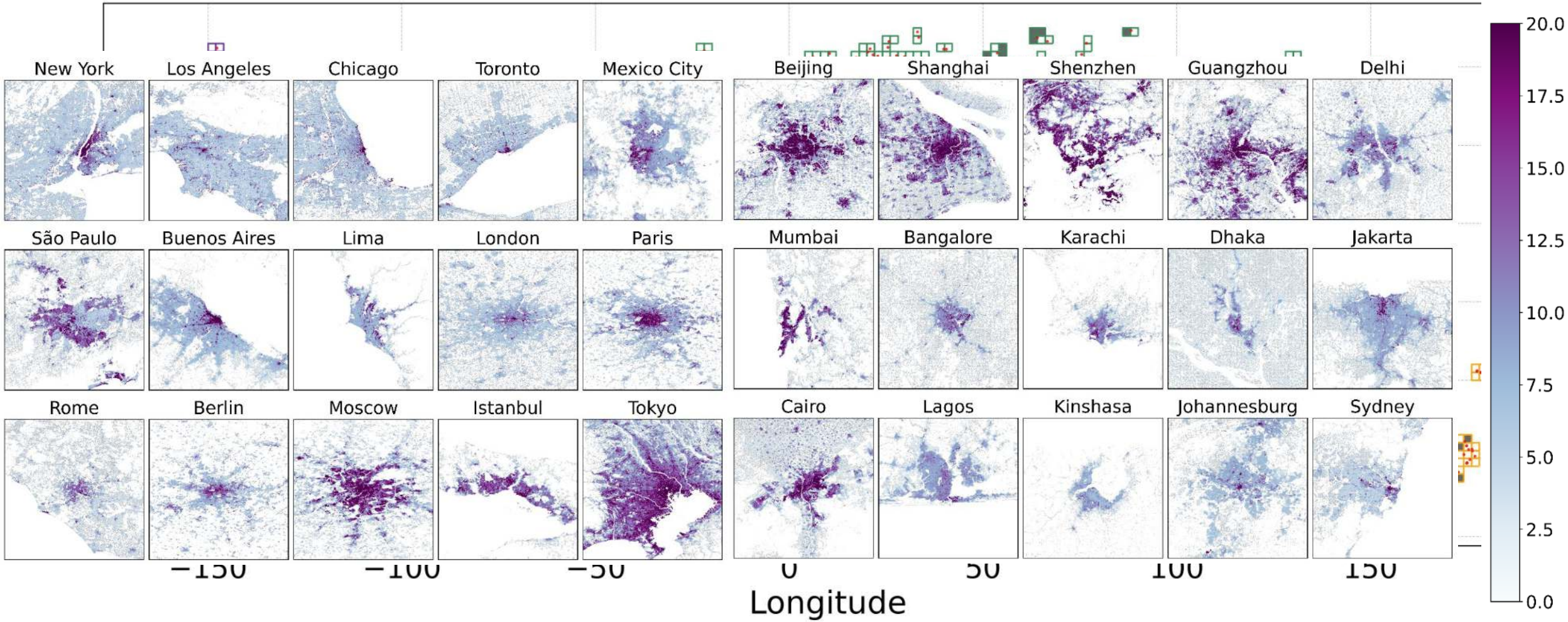
Cities/Locales Processed:
11,103

Total Building Footprints:
~2.38 Billion

Target Resolution:
100-Meter Grid

Model-Ready Data for Advanced Simulations

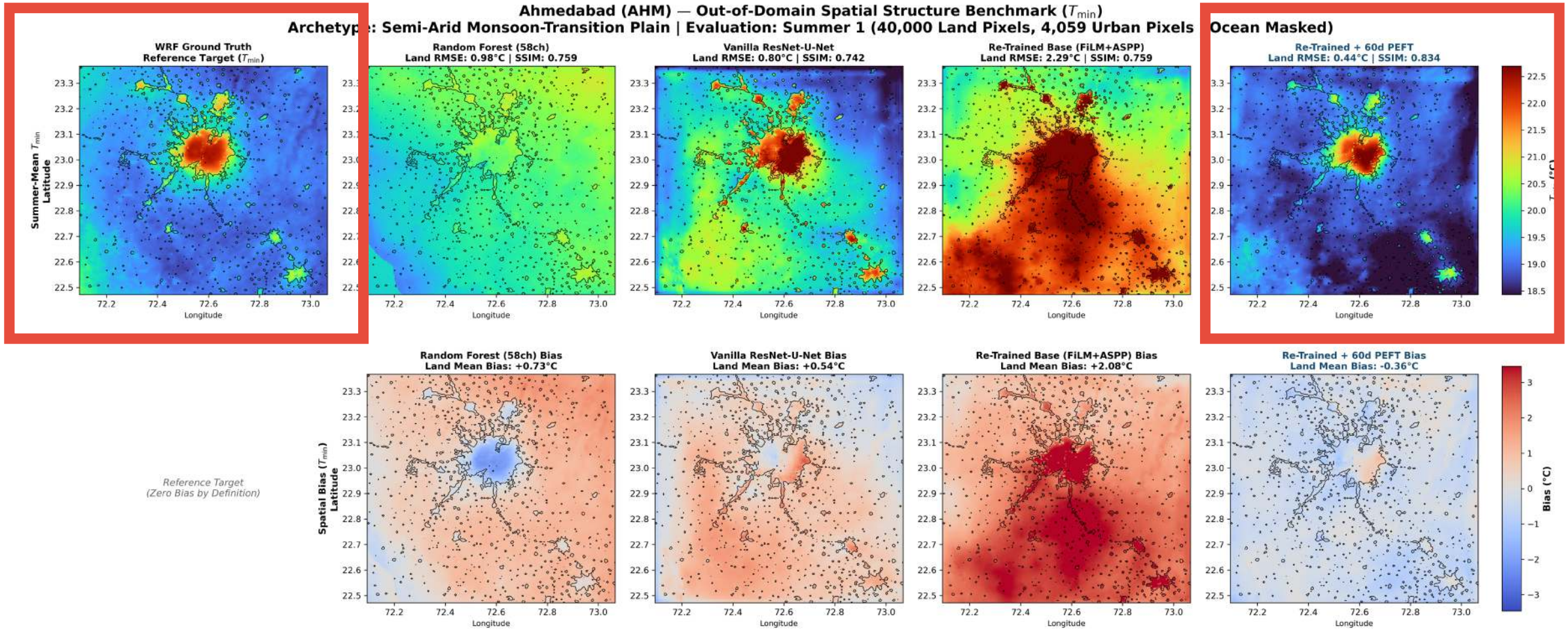
UCPs for (nearly) all cities



AI-based downscaling

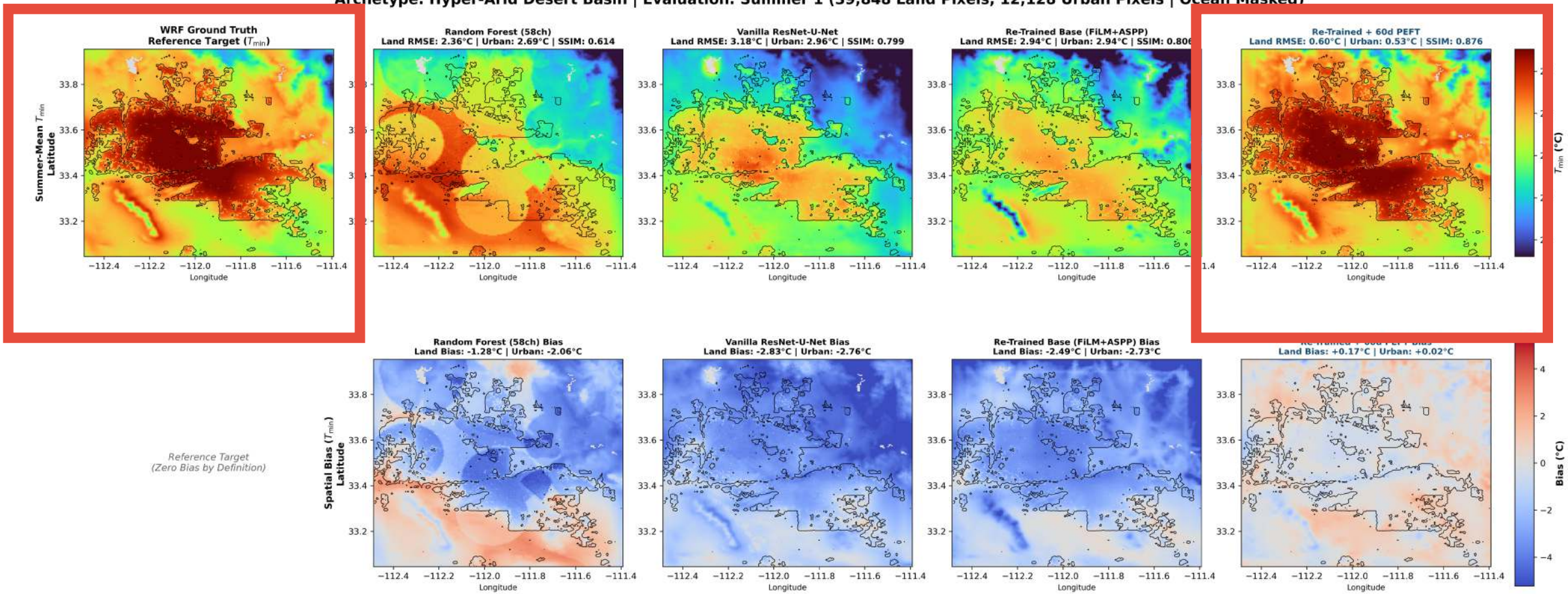
- U-Net Model trained on Residual from coarse input
  - Predicts fine-scale spatial residual offsets (target minus coarse base) on top of interpolated coarse dynamic inputs.
  - Feature-wise Linear Modulation (FiLM) used to maximize utility of time-invariant, but spatially detailed urban parameters
- Predictors:
  - Dynamic: 100 km wind, 2-meter temperature, surface pressure, soil
  - Static: UCPs, topography, land mask
  - Land use index

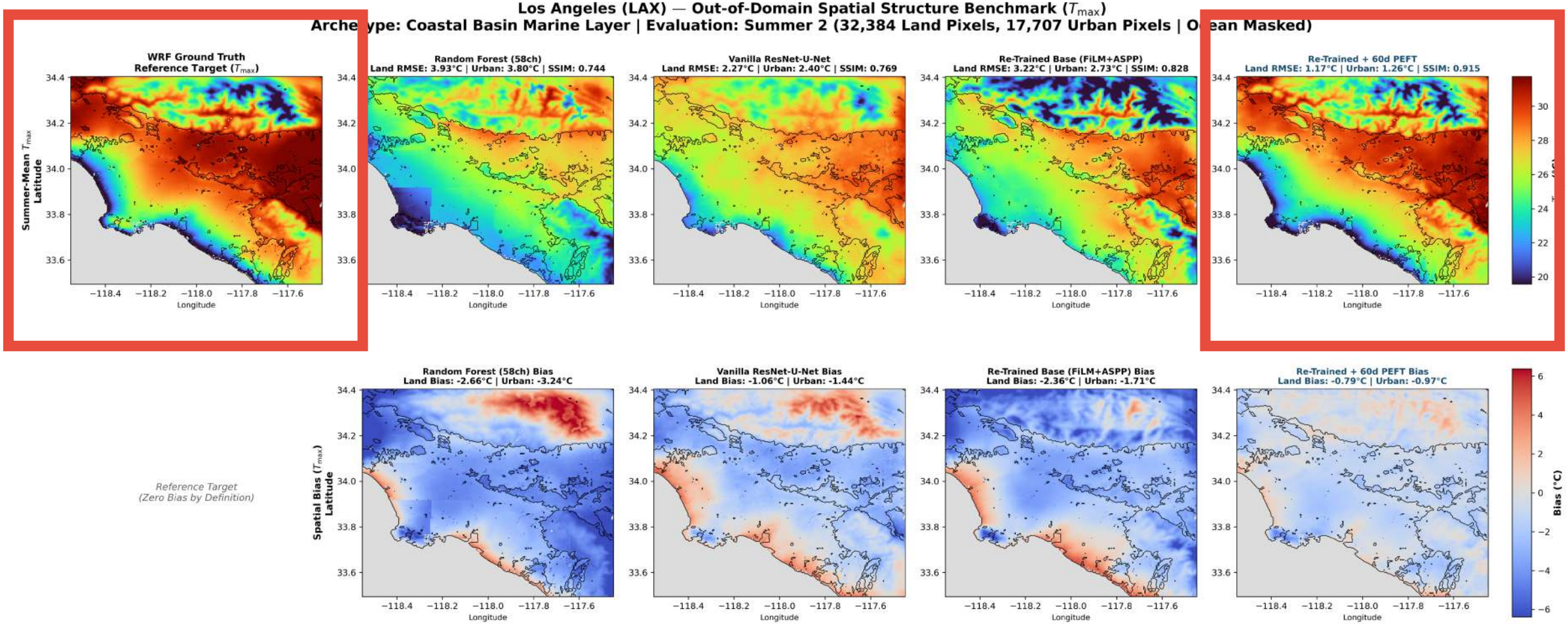
Preliminary results



Source: *Ortiz, L., Jain, M. A global approach to neighborhood scale weather modeling (In preparation)*

Phoenix (PHX) — Out-of-Domain Spatial Structure Benchmark (T_{min})
Archetype: Hyper-Arid Desert Basin | Evaluation: Summer 1 (39,848 Land Pixels, 12,128 Urban Pixels | Ocean Masked)




Los Angeles (LAX) — Out-of-Domain Spatial Structure Benchmark (T_{max})
Archetype: Coastal Basin Marine Layer | Evaluation: Summer 2 (32,384 Land Pixels, 17,707 Urban Pixels | Ocean Masked)
WRF Ground Truth
Reference Target (T_{max})
Random Forest (58ch)
Land RMSE: 3.93°C | Urban: 3.80°C | SSIM: 0.744
Vanilla ResNet-U-Net
Land RMSE: 2.27°C | Urban: 2.40°C | SSIM: 0.769
Re-Trained Base (FiLM+ASPP)
Land RMSE: 3.22°C | Urban: 2.73°C | SSIM: 0.828
Re-Trained + 60d PEFT
Land RMSE: 1.17°C | Urban: 1.26°C | SSIM: 0.915
Summer-Mean T_{max}
Latitude
Longitude
Reference Target
(Zero Bias by Definition)
Random Forest (58ch) Bias
Land Bias: -2.66°C | Urban: -3.24°C
Vanilla ResNet-U-Net Bias
Land Bias: -1.06°C | Urban: -1.44°C
Re-Trained Base (FiLM+ASPP) Bias
Land Bias: -2.36°C | Urban: -1.71°C
Re-Trained + 60d PEFT Bias
Land Bias: -0.79°C | Urban: -0.97°C
Spatial Bias (T_{max})
Bias (°C)

Final remarks

[1] Cities are important: They affect large swaths of people and infrastructure

[2] Cities affect climate: Their unique properties lead to unique meteorological conditions

[3] Modeling urban climates is expensive: Spatial resolution, specialized physics modules, specialized data needs

[3.1] New work is addressing these challenges: uMORPH brings model-required parameters to most of the world

[3.2] Recent advances are creating a framework to model (cheaply for now, VERY cheaply in the future) urban climates at scale.

[4] Opportunities to inform urban planning, urban emergency response by helping understand how changes in urban environment lead to changes in local hazards.